WESTFIELD CASE

PART 3

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Recap: Predicting Service Needs

- Planning capacity requires understanding how long clients are likely to remain in care
- We built two OLS models to predict the number of appointments per child from structured visit and intake data
- Model 2 with patient type, modality, and their interaction achieved an **R^2 value of 0.924**
  - Meaning this regression model can explain 94% of variance and is a strong overall fit to historical data
- However, R2 is a population-level statistic and does not guarantee accurate prediction for individual children

OLS Regression Results

Dep. Variable:	num_appt	R-squared:	0.942
Model:	OLS	Adj. R-squared:	0.942
Method:	Least Squares	F-statistic:	3630.
Date:	Mon, 03 Mar 2025	Prob (F-statistic):	0.00
Time:	20:53:17	Log-Likelihood:	-2601.7
No. Observations:	1800	AIC:	5221.
Df Residuals:	1791	BIC:	5271.
Df Model:	8		
Covariance Type:	nonrobust		

	coef	std err	t	P>|t|	[0.025	0.975]
Intercept	10.0050	0.073	137.450	0.000	9.862	10.148
C(pat_type)[T.2]	1.9900	0.103	19.332	0.000	1.788	2.192
C(pat_type)[T.3]	3.9950	0.103	38.809	0.000	3.793	4.197
C(modality)[T.G]	0.9200	0.103	8.937	0.000	0.718	1.122
C(modality)[T.V]	-0.0200	0.103	-0.194	0.846	-0.222	0.182
C(pat_type)[T.2]:C(modality)[T.G]	2.1300	0.146	14.631	0.000	1.844	2.416
C(pat_type)[T.3]:C(modality)[T.G]	9.1250	0.146	62.680	0.000	8.839	9.411
C(pat_type)[T.2]:C(modality)[T.V]	0.6250	0.146	4.293	0.000	0.339	0.911
C(pat_type)[T.3]:C(modality)[T.V]	2.1450	0.146	14.734	0.000	1.859	2.431

Omnibus:	1.089	Durbin-Watson:	1.897
Prob(Omnibus):	0.580	Jarque-Bera (JB):	1.133
Skew:	0.058	Prob(JB):	0.568
Kurtosis:	2.957	Cond. No.	13.9

Structured Data Only Tells Part of a Client's Story

- Structured intake data includes only a limited set of coded variables
- But the number of sessions a child ultimately needs is influenced by many factors that are difficult to represent in simple fields or through linear models
- These may include:
  - Variations in clinical judgement
  - Family factors (e.g., parent engagement in and out of sessions, stressors at home)
  - Attendance barriers (e.g., family car broke down, child becomes sick)
  - Response to therapy (e.g., some may lose interest and not engage with therapy activities)
  - Functional concerns

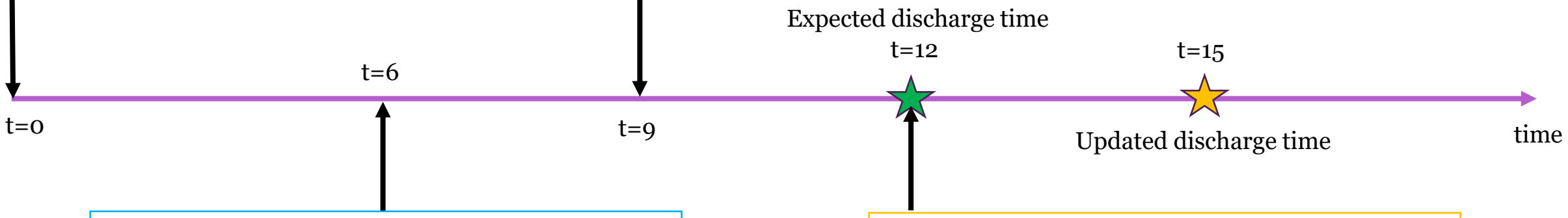
Service Needs Can Change Over Time

Referral assessment (t=0)

- Client is initially assessed as a mild case (P1)
- Estimated need: **10 sessions**

Halfway-through TX, a change in family circumstances (t = 9)

- A new sibling is born, making it more difficult for parents to practice strategies at home
- The clinician begins to observe **slower progress between sessions**



Service begins (t = 6)

When capacity becomes available, the client starts service.

At the initial clinical assessment, the clinician finds that needs have increased since referral:

- Re-assessed as a moderate case (P2)
- Updated estimate: **12 sessions**

Reassessment at expected discharge (t = 12)

At the scheduled reassessment, the clinician determines that the client would benefit from additional support to consolidate progress.

An extra block of therapy is added.

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Clinician Caseload Management

- Not only are clients heterogeneous in their needs, but clinicians are also different in their decision-making
- At Westfield, clinicians are responsible for managing their own caseloads
  - There are **maximum service limits**, but actual discharge timing is not fixed
  - In practice, clinicians decide whether a child continues or is discharged based on clinical judgment at re-assessment points
- Sources of variation:
  - Risk tolerance (e.g., some clinicians may extend care to increase certainty that client has made sufficient progress)
  - Clinical experience and specialization

Similar Client Profiles, Different Outcomes

- Two children with identical intake profiles, assigned to different clinicians, may receive very different numbers of sessions
- Example:
  - **Jane** has been working at Westfield for over 5 years, and approximately 70% of her caseload consists of Motor Speech clients. Through repeated experience with this population, she has developed efficient assessment and treatment strategies and may be able to achieve progress in fewer sessions.
  - **Cathy** has been at Westfield for 1 year, and only about 15% of her caseload consists of Motor Speech clients. With less experience treating this specific client group, she may require more time to assess needs, refine treatment plans, and achieve the same outcomes.

Client A

Priority: P1
Age: 2 years old
Gender: Male
Referral Reason: Motor Speech
Received: 10 sessions

Clinician: Jane, SLP

Client B

Priority: P1
Age: 2 years old
Gender: Male
Referral Reason: Motor Speech
Received: 15 sessions

Clinician: Cathy, SLP

Hidden Signals in Clinical Documentation

- Clinical notes contain more than a summary of what happened in a session
- They often capture signals that are not available in structured intake data, such as:
  - response to therapy
  - level of cueing or support needed
  - caregiver follow-through at home
  - emerging family or attendance barriers
  - readiness to progress, pause, or discharge
- These signals can help us categorize clients based on progress over time and enable more **dynamic capacity planning and scheduling**

What a Clinical Note Contains

Progress Note – Child A, Session 4

- Mom reported child is more understandable at home after practicing at home
- Child will be starting JK in the fall at School Y
- Client targeted 3-word phrase combinations at the phrase level. Required minimal verbal cuing to initiate. Accuracy approximately 7/10 trials.
- Client demonstrated spontaneous use of target structures in play-based context without prompting.
- Goals advanced to sentence level for next session. Parent reports increased use at home.

🗣️ **Subjective/Reported Progress at Home**

Mom reports content with progress with practice being carried over at home

🎯 **Performance on therapy targets**

Child achieved good accuracy on 3-word phrases

🤝 **Level of support needed**

Child only needed minimal cueing to do activity

📈 **Readiness for next stage/care planning**

Goals were advanced, indicating readiness for more complex speech tasks.

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Operationally, What Could This Note Signal?

- Client is doing well, does that mean they are ready for discharge?
  - Maybe not… Client is experiencing a really good rate of progress and stopping them at this point could interrupt that growth
- Client is starting JK this fall
  - We need to be ready to discharge them before the next school year
  - Parents may seek more frequent or higher intensity service to capitalize on the limited time they have left

Progress Note – Child A, Session 4

- Mom reported child is more understandable at home after practicing at home
- Child will be starting JK in the fall at School Y
- Client targeted 3-word phrase combinations at the phrase level. Required minimal verbal cuing to initiate. Accuracy approximately 7/10 trials.
- Client demonstrated spontaneous use of target structures in play-based context without prompting.
- Goals advanced to sentence level for next session. Parent reports increased use at home.

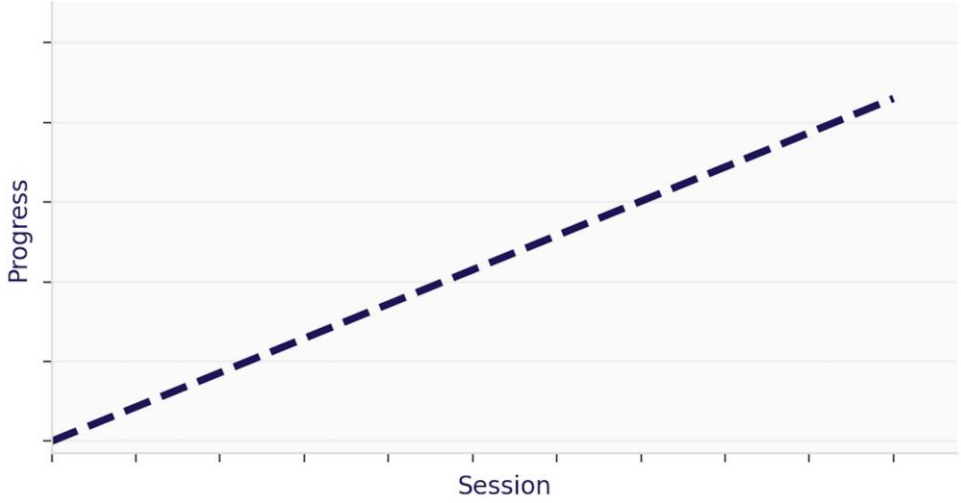
A single note alone may not be a reliable signal

- A single note may reflect a good or bad session whereas consecutive notes reveal the underlying trajectory

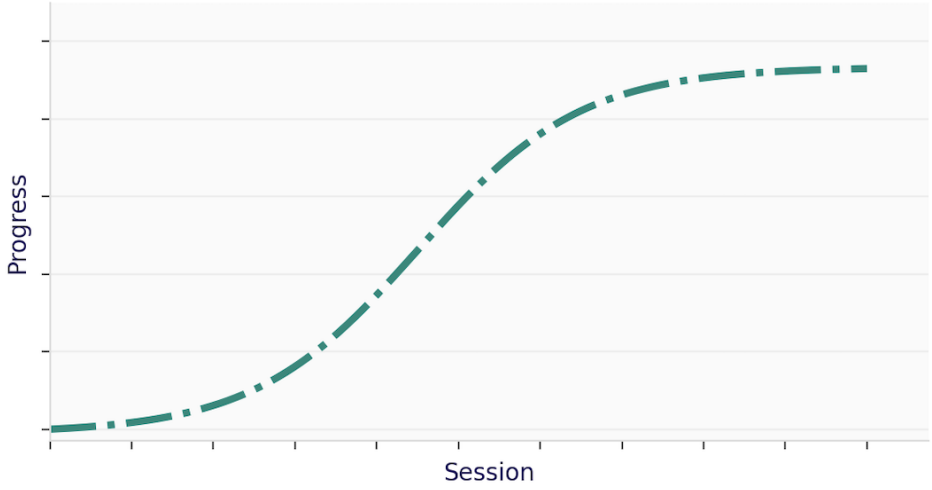
Session 1	Session 2	Session 3	Session 4
Client is progressing well and used target words independently in play. Parent reports strong carryover at home.	Client required some prompting throughout the session and showed limited accuracy.	Progress remains slow. Client was dysregulated and unable to sustain participation in structured tasks.	Minimal improvement noted. Parent reports difficulty completing home practice.

Building a Longitudinal Signal

- What if we could translate a sequence of progress signals into one longitudinal signal about a client?
- If we score each note based on how much relative progress has been made, we could visualize a client's trajectory through treatment



Linear Progress: Client improves at same rate session after session



Slow-Fast-Slow Progress: Client starts slow, has a burst of progress mid-way, then plateaus near the end

Operational Value of Quantifying Clinical Progress

1. Support more informed clinical decisions

- Clinicians are focused on individual sessions and may not always see the broader trajectory across weeks or months
- Summarizing progress trends over time can help them recognize whether a child is accelerating, plateauing, or regressing, and adjust goals or treatment plans accordingly

2. Detect diminishing returns in therapy

- Helps identify possible plateaus, where additional sessions may be producing smaller incremental benefit

3. Enable better timing of reassessment or intervention

- A stagnant or declining pattern across notes can be used as an early signal to review the case.

Challenges of Using Progress Signals from Clinical Notes

- These signals of progress and learning are stored in **free-text notes** which makes them hard to analyze systematically compared to structured data
- Progress is usually described **qualitatively rather than numerically**, which makes it difficult to determine how much progress has been achieved
- If each client has a note per session, clinics will have tens of thousands of notes
  - Making it nearly impossible to manually review and analyze each note

To make these signals usable at scale, we need automated approaches for extracting meaningful insights from large volumes of unstructured clinical notes

TEXT-MINING APPROACHES

Extracting Planning Signals from Clinical Notes

- There are two approaches we explored:
  1. **Rule-based NLP**: use predefined keywords, dictionaries, and patterns
  2. **LLMs**: use context-aware language models to interpret meaning in notes
- Broadly, these approaches differ in:
  - **Interpretability** — how easy it is to understand why/how a signal was extracted
  - **Control** — how precisely the method can be designed or constrained
  - **Flexibility** — how well it handles varied wording and note styles
  - **Power** — how well it captures context, nuance, and complex patterns

Traditional Rule-Based NLP: Matching Words and Patterns

- A simple NLP approach is to search notes for predefined keywords or regex patterns and use them as signals of progress
- We can define a domain-specific vocabulary of phrases associated with progress, such as:
  - Positive: progressing, doing well, made gains, consistent
  - Negative: no improvement, having difficulty, not progressing
- The system scans each clinical note and applies these patterns to detect whether any predefined phrases appear

Sample note by Clinician A

Client is progressing in goal X...

Doing well with new tasks

Having difficulty with goal Y...

Accuracy of 6/10 for goal Z

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

“Sentiment Analysis” Approach for Progress Scores

- Each matched phrase contributes a positive or negative score to the note as a whole
- The total score becomes the estimated progress sentiment for that session
- Tracked over sessions, these scores can reveal a client’s trajectory: improving, plateauing, or declining
- This is one of the simplest ways to turn **unstructured text** into **structured data**

Sample note by **Clinician A**	
Client is progressing in goal X…	+1
Doing well with new tasks	+1
Having difficulty with goal Y…	-1
Accuracy of 6/10 for goal Z	+1

Estimated Progress Sentiment: +2

Challenges Encountered

Keywords:

"progressing"
"doing well"
"made gains"
"consistent"
"_% accuracy" >= 50
"no improvement"
"having difficulty"
"not progressing"
"_% accuracy" < 50

Sample note by **Clinician A**
Client is progressing in goal X…
Doing well with new tasks
Having difficulty with goal Y…
60% accuracy for goal Z

Estimated Progress Sentiment: +2

Sample note by **Clinician B**	
No longer having difficulty in goal X	-1
Responding well to new tasks	0
Making gains with goal Y…	+1
Accuracy of 7/10 for goal Z	0

Estimated Progress Sentiment: 0

True Progress Sentiment: +4

The negation changes the meaning here

No rule to capture these expressions yet

Takeaway: Clinical Notes Require Context-Aware NLP

- Rule-based NLP can detect surface-level words and phrases, but it does not truly understand meaning
- **Language variation creates a high rule burden:** the same concept may be described differently across clinicians, requiring hundreds of regex rules
- Negation, nuance, and mixed signals make pattern matching unreliable
- Meaning depends on context, not just keywords
- This makes rule-based approaches brittle and hard to generalize

For robust clinical note interpretation, we need methods that understand context rather than simply match patterns

Large Language Models for Clinical Text Analysis

- Rather than build a traditional NLP pipeline from scratch, we explore whether LLMs can extract meaningful signals directly from free-text clinical notes
- LLMs can interpret **context, nuance, and varied phrasing** that are difficult to capture using rules or structured fields alone
- Specifically, generative LLMs (e.g., ChatGPT) can be prompted to perform classification, extraction, and summarization without re-training

Example zero-shot prompt:

```
You are a clinical assistant. Read the referral note below and extract the following as JSON:
- primary_diagnosis (string)
- recommended_services (list)
- urgency (High / Medium / Low)

Note: "3 y.o. male with suspected gross motor delay. Please assess for OT and SLP. Parents report regression
over past 3 months."
```


UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Importance of Prompt Engineering

- The quality of LLM output is highly sensitive to how the task is described. Effective prompts share common patterns

Technique	Example	Why It Helps
Role assignment	"You are a pediatric speech-language therapist who is reviewing the following note to…"	Anchors the model's vocabulary and clinical reasoning style
Output format specification	"Return a JSON object with keys: targeted goals, a progress score between 1-10"	Forces structured output that is easy to work with
Confidence elicitation	"Rate your confidence 1-5 and explain any uncertainty"	Flags low-confidence outputs for human review

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Few-Shot Prompting

- A zero-shot prompt asks the model to classify a note with no examples. This works for general language but struggles with clinical text, where the same concept is expressed in dozens of different ways across clinicians
- Few-shot prompting solves this by providing 2-5 labelled examples before the new note, anchoring the model to your specific labelling conventions

Example few-shot prompt:

```
You are a clinical assistant. Read the referral note below
and extract the following as JSON:
- primary_diagnosis (string)
- recommended_services (list)
-  urgency (High / Medium / Low)


_____
Here are some examples to learn from:
1. Note: “Client achieved 8/10 accuracy. Goals advanced to
sentence level. Parent reports carry-over at home.”
→ Label: Progress = High | Discharge Proximity = Near

2. Note: “Client struggled with target. Required maximum
cueing. No carry-over reported.”
→ Label: Progress = Low | Discharge Proximity = Far
____

Note: "3 y.o. male with suspected gross motor delay.
Please assess for OT and SLP. Parents report regression
over past 3 months."
```


UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Chain-of-Thought: Makes the Model Think Out Loud

- Chain-of-thought (CoT) prompting instructs the model to **reason step-by-step before producing its final answer**, surfacing the logic behind each classification
- Why CoT can be valuable in a clinical context
  - **Auditability:** Clinicians and managers can read the model's reasoning, not just its label
  - **Error diagnosis:** When the model gets it wrong, the reasoning step reveals where, enabling targeted prompt refinement rather than guesswork

Chain-of-Thought Instruction:

```
First, list all progress signals
mentioned.

Second, identify any discharge-
relevant signals such as goal
advancement, family carry-over, or
external factors.

Third, note any contradictions.

Finally, rate discharge readiness as
Near / Moderate / Far and explain why
```

Hallucination Risk

- LLMs can generate plausible but factually incorrect outputs, a phenomenon known as hallucination
- In clinical settings, hallucinations are especially undesirable because incorrect outputs may influence care decisions
- **Common forms of hallucination in clinical NLP:**
  - Fabricating clinical details not mentioned in the note (e.g., inventing a diagnosis)
  - Misattributing progress signals to the wrong goal or session
  - Generating confident classifications even when the note is ambiguous or incomplete
- **Mitigation strategies:**
  - Ask the model to cite evidence from the note before classifying
  - Request confidence scores and flag low-confidence outputs for human review
  - Use chain-of-thought prompting to expose reasoning before the final label

Validation Method: Human-in-the-loop

- LLM outputs must be validated before being used to inform clinical or operational decisions
- Human-in-the-loop (HITL) validation involves clinicians reviewing model outputs on a sample of cases

A practical HITL workflow:

1. Model classifies a batch of clinical notes (e.g., progress = High / Medium / Low)
2. A clinician reviews a random stratified sample and records their own label
3. Model and human labels are compared to compute agreement metrics
4. Disagreements are examined to identify systematic errors and improve the prompt or model

Agreement Metric: Confusion Matrix

of Low Progress scores predicted = 85

Accuracy for "Low Progress" = 85/105

True/Pred	Low Progress	Moderate Progress	High Progress	Row Total
Low Progress	85	14	6	105
Moderate Progress	5	32	19	56
High Progress	0	8	32	40
Col Total	90	54	57	201

True # of Low Progress scores = 105

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Agreement Metric: Confusion Matrix

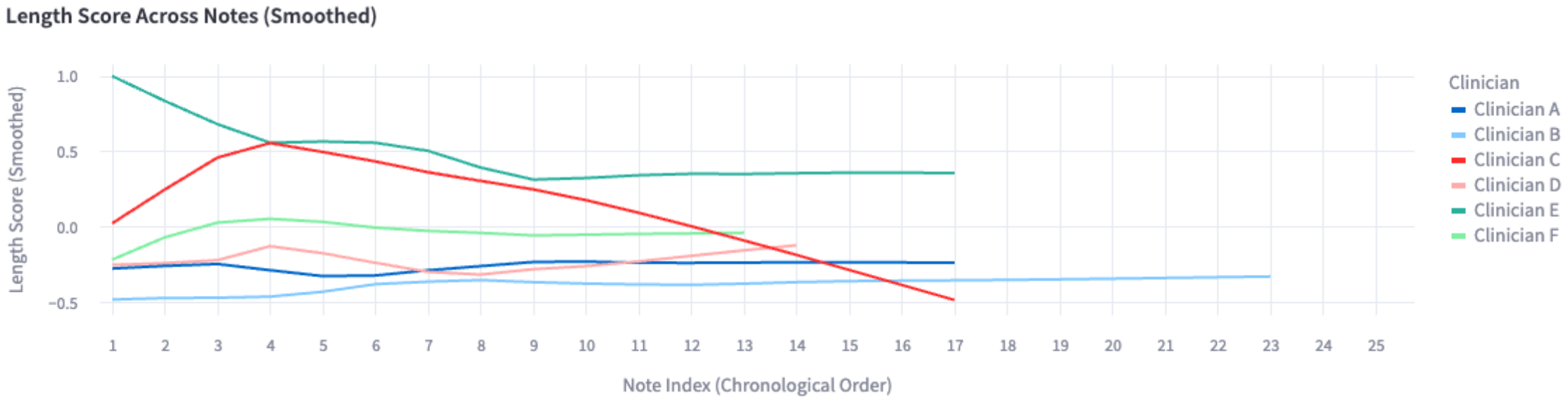
- Using accuracy alone can be misleading when class distributions are imbalanced (e.g., we have a lot of high or moderate progress, but not many low)
- The confusion matrix helps us learn:
  - Which classes the model confuses with each other? – e.g., is moderate being mistaken often for high or low progress?
  - **Recall per class** – of all the truly High progress notes, how many did the model actually catch? (The row diagonal divided by the row total)
    - e.g., Recall for Low = 85/105 = 0.81. There were 105 truly Low cases, the model identified 85 of them
  - **Precision per class** – of all the notes the model called High progress, how many actually were? (The column diagonal divided by the column total)
    - e.g., Precision for Low = 85/90 = 0.94. The model predicted Low 90 times but only 85 of them were truly Low

Bias in Training Data

- Documentation can vary significantly between clinicians
  - Some may write with a more positive/negative tone
  - Some may write every detail while others write the main points needed for the file
- If models are trained on notes from a subset of clinicians or client populations, outputs may not generalize fairly
  - If we train an LLM to measure progress in notes that only target Motor Speech Delays, it may not perform accurately on other speech-therapy disorders
- Similarly, we want an LLM to be able to generalize across the various programs offered (SLP, OT, PT)

Document Length Over Time

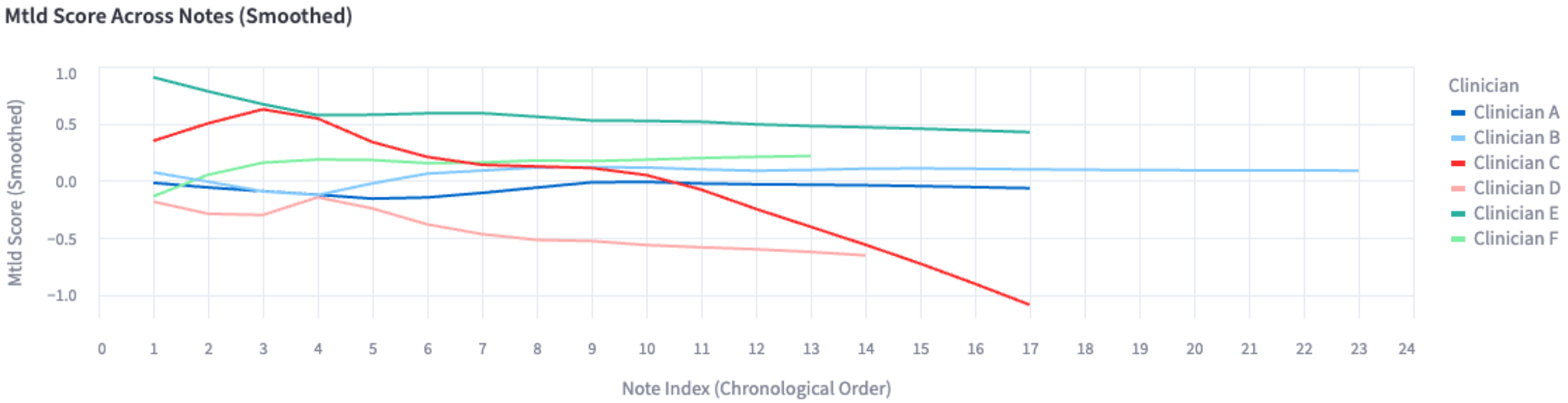
- We can see these differences in clinical documentation by seeing how length of their notes change over time
- Clinician E writes long notes in earlier sessions, then gradually drops off. However, they still write on average more than the other clinicians across all sessions.



UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Lexical Diversity

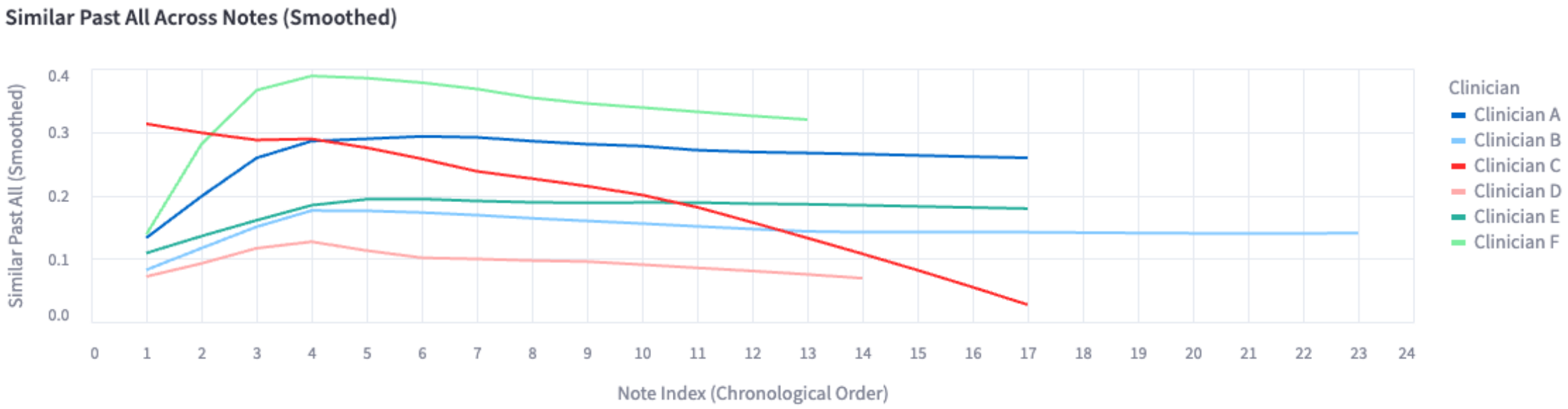
- Length is not always a strong indicator of note informativeness, as it could mean a lot of repetition, so we can score it on the diversity of vocab used
- **MTLD (Measure of Textual Lexical Diversity)** is a score that measures how varied the vocabulary in a text is by calculating the average length of consecutive word sequences before the proportion of unique words drops below a set threshold — longer sequences before that drop indicate richer lexical diversity.



UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Similarity Scores to the Previous Note

- Since clinicians write progress notes each session, they often copy information from past notes (e.g., same list of goals targeted) to save documentation time
- High similarity to prior notes could potentially be less informative, since a note that is nearly identical to previous sessions might not indicate much about progress
- But clinicians with high similarity scores could also indicate consistency between how they structure their notes, making them clear cases for training an LLM



Summary: Clinical Notes to Planning Signals Workflow

Clinical Notes
Pass a client's session notes to the LLM

→

Signal Extraction
Ask the LLM to score the progress made between each consecutive note on some numeric scale

→

Build Vector of Scores
We can represent the scores a vector
e.g., **[3, 2, 1, …., 1]** where 3 is the relative progress between session 1 and 2

↓

Client Trajectory
We plot the cumulative progress over time to show the client's learning pattern

↓

Decision Support
Use the curve for predictive and prescriptive modeling

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

PLANNING USING LEARNING CURVES

Clustering Clients by Trajectories

Why cluster?

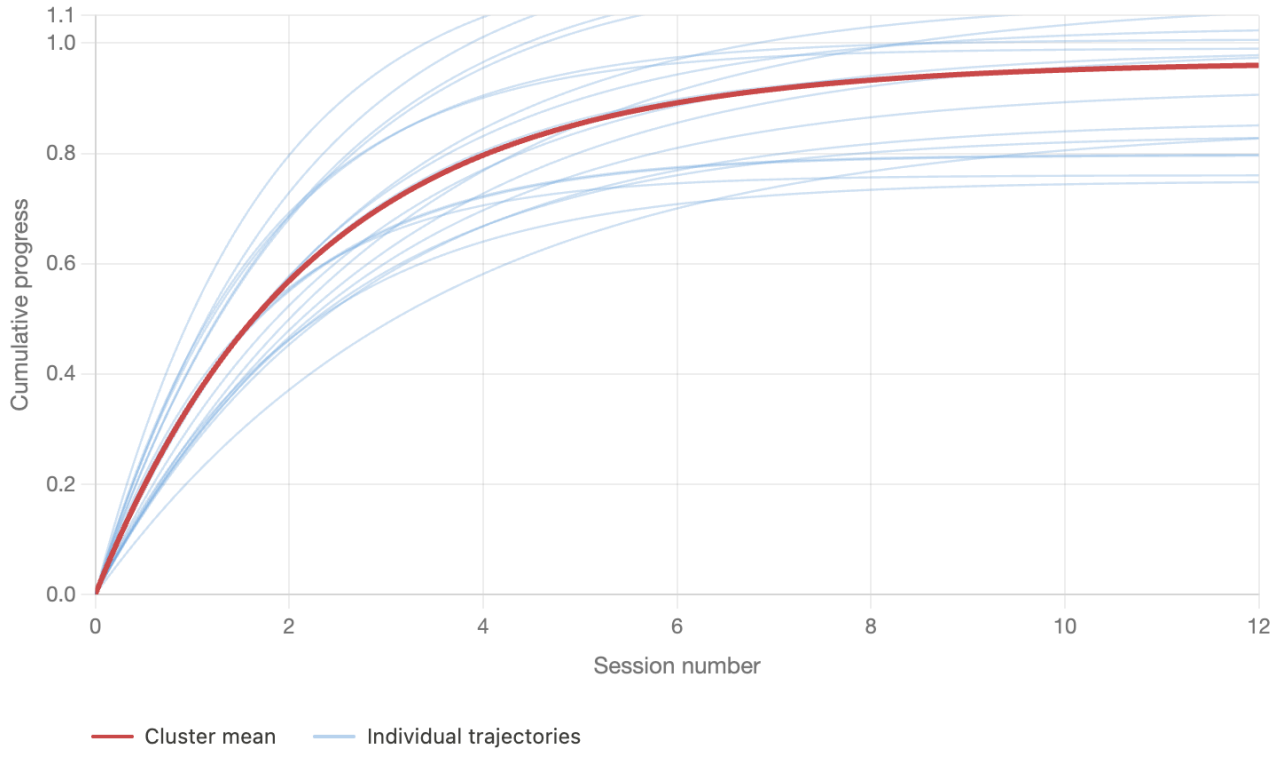
Individual learning curves vary widely. Clustering reveals systematic patterns shared across clients → enabling population-level planning.

How we cluster

We use the full sequence of progress delta scores as a multivariate feature vector, then apply trajectory-based clustering (e.g. k-means).

What clusters give us

Each cluster has a characteristic trajectory shape. In pediatric rehabilitation, clients typically progress at a measurable rate before eventually plateauing once they have achieved their functional goals or reached the limits of their current capacity

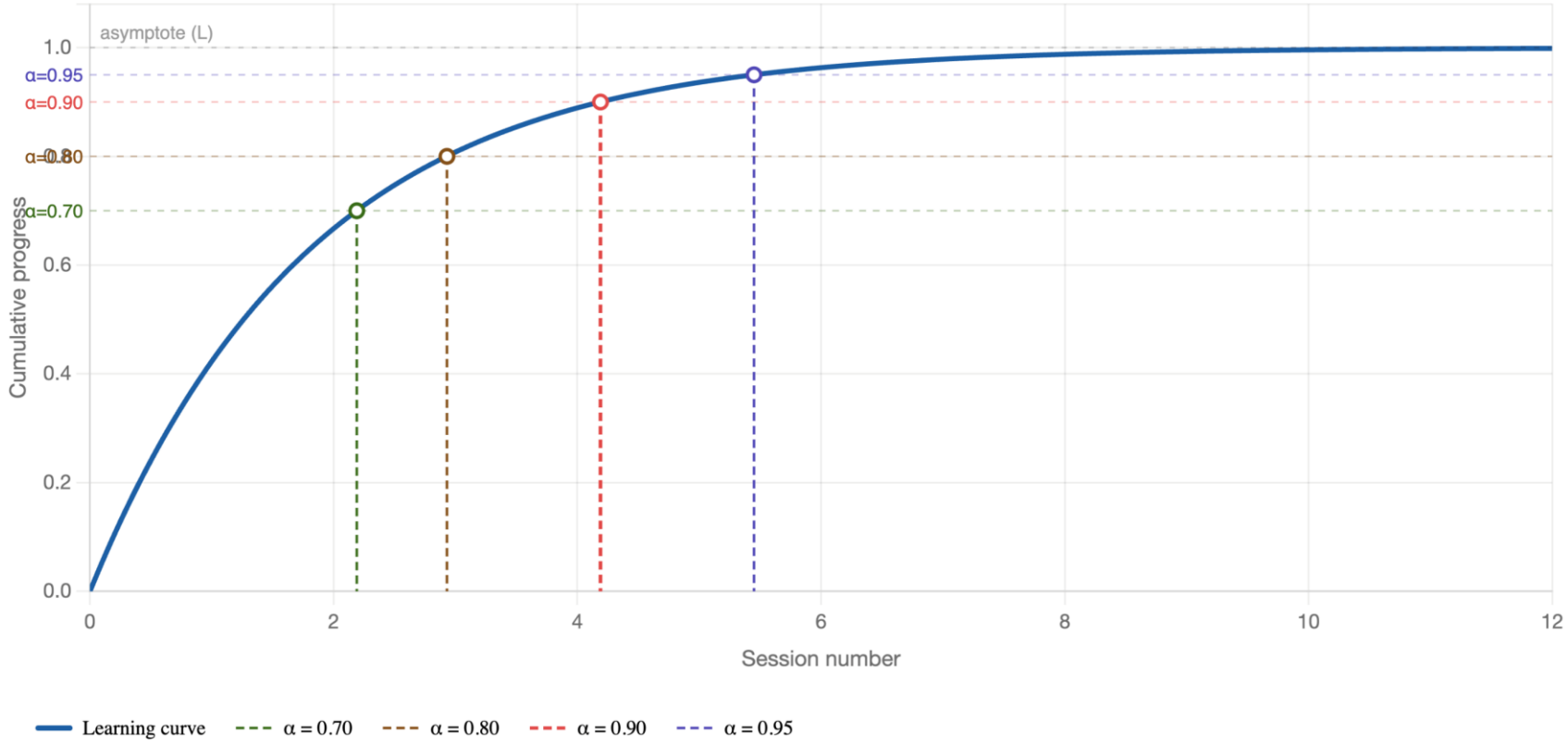


Applying a Stopping Rule

- Many sequential processes yield diminishing returns over time
- Continuing to provide sessions past the point of meaningful gain can be wasteful but stopping too early forfeits real benefit
- The solution is a **stopping rule**: a principled criterion that defines when enough progress has been made
- **In clinical rehab, this becomes: when should we reassess or discharge?**

Defining the Stopping Point

- **Introduce alpha (α) as a threshold**
  - Rather than waiting for full convergence (which never technically occurs on an asymptotic curve), we define alpha in (0,1) as the fraction of total expected progress we consider sufficient
  - Setting alpha = 0.90 means 90% of potential gain is enough to trigger a reassessment decision



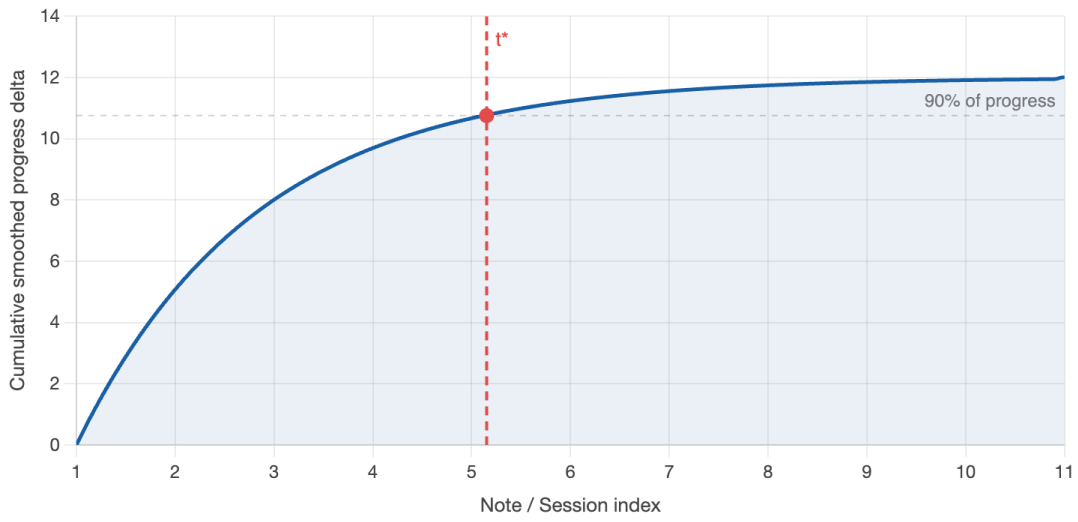
UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Using a 90% Threshold to Define a Reassessment Point

Single client

Individual learning curve (t*)

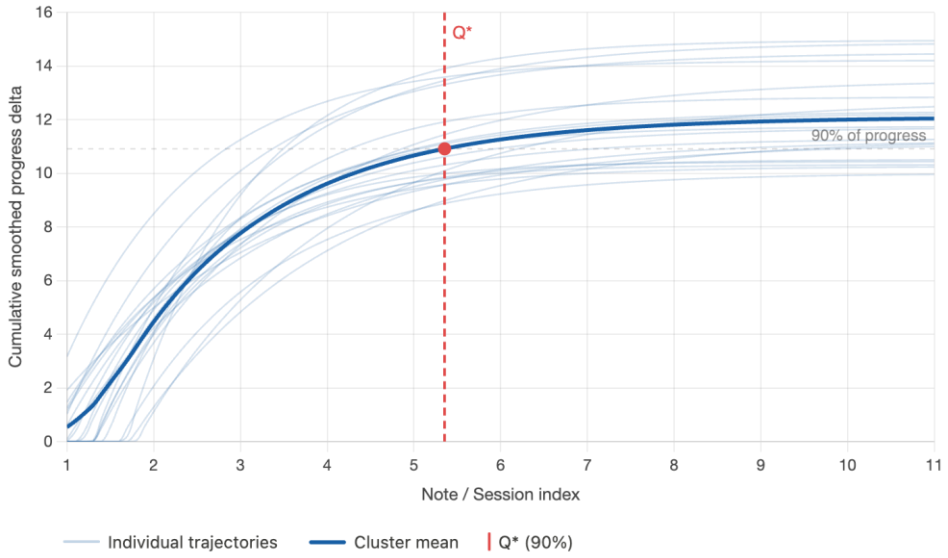
Each client follows their own growth path from (0, 0) toward an asymptote. The individual 90% plateau point is their personal t*.



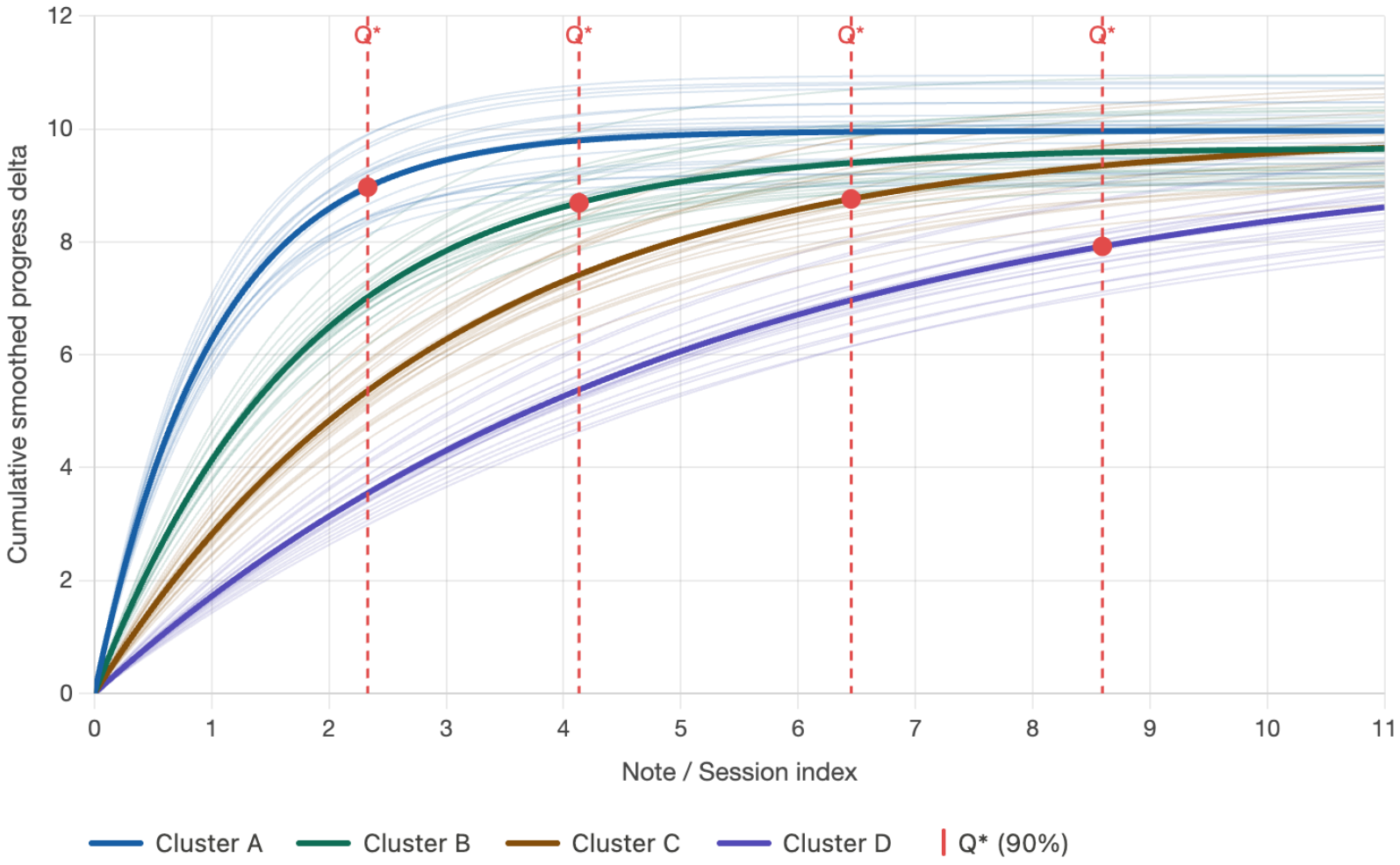
Cluster

Cluster mean curve (Q*)

Averaging across all clients in a cluster gives a smooth mean trajectory. Q* is defined as the session where the cluster mean reaches 90% of its progress range.

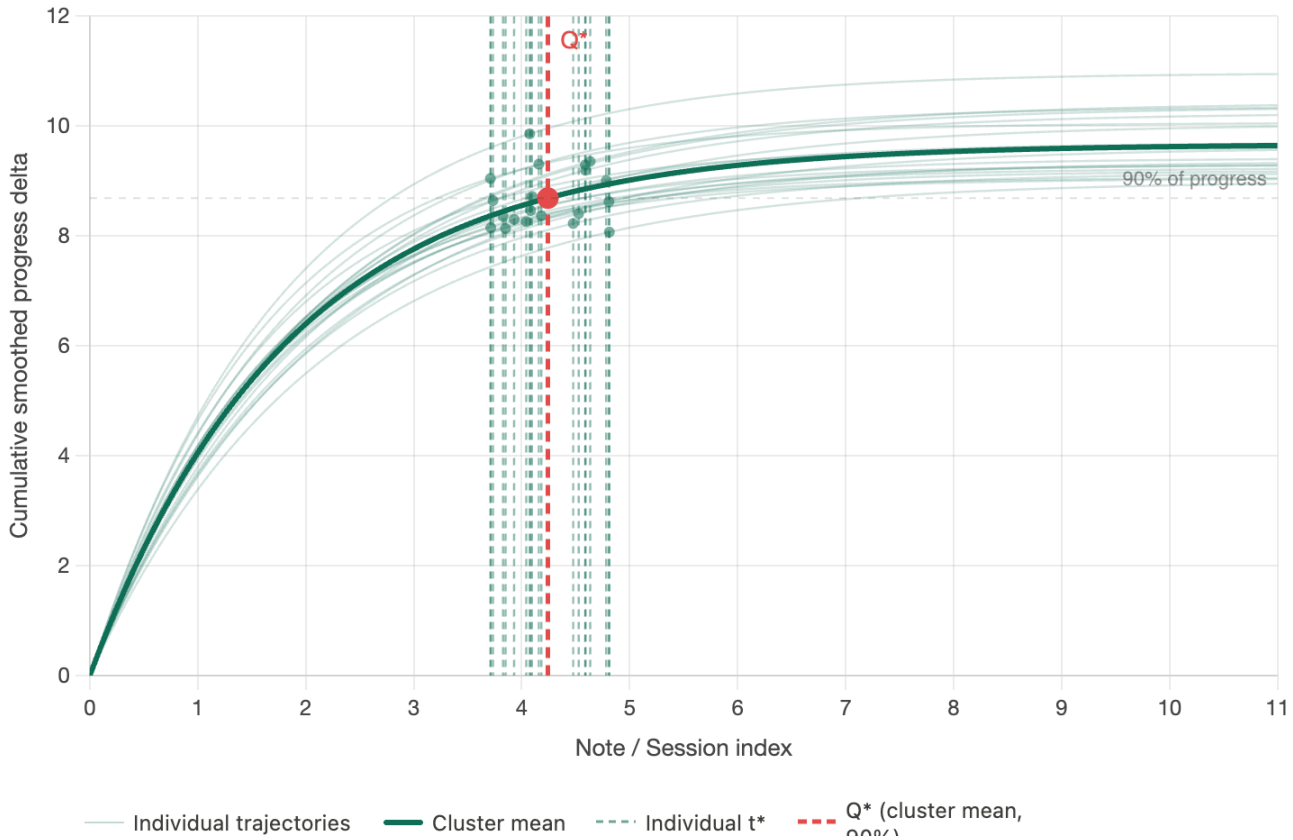


Q* Across Many Clusters



UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Individual t* vs Cluster Q*



Q* is a group property

It represents the session where the average client in the cluster reaches 90% of their total progress.

t* varies within a cluster

Individual clients can plateau earlier or later than Q*, even within the same trajectory group.

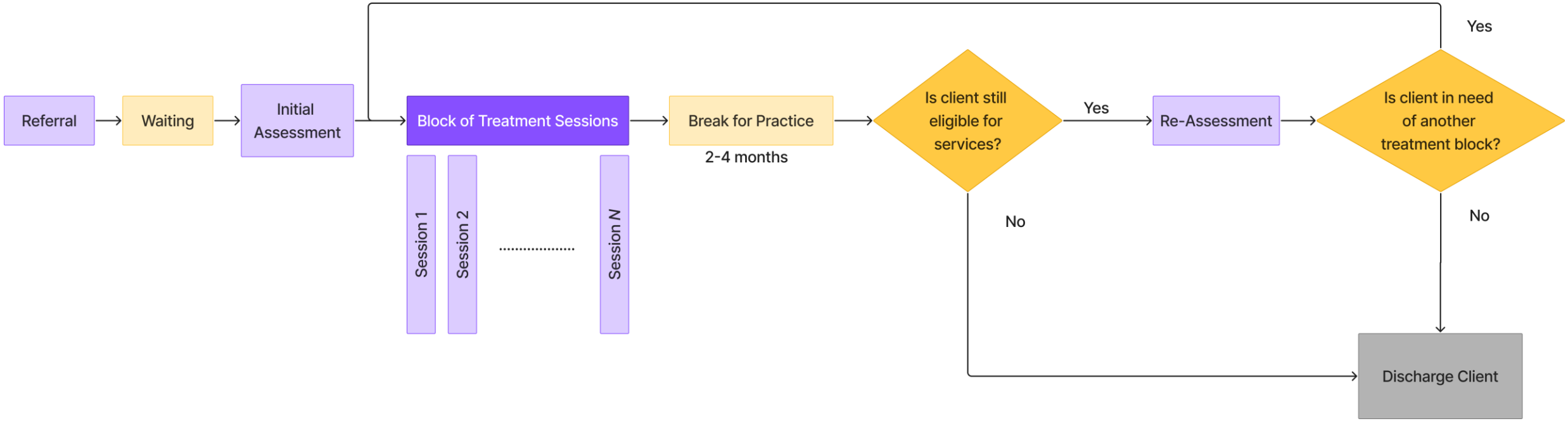
Spread matters for policy

The wider the t* distribution within a cluster, the more uncertain our reassessment timing will be.

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Multi-Appointment Care

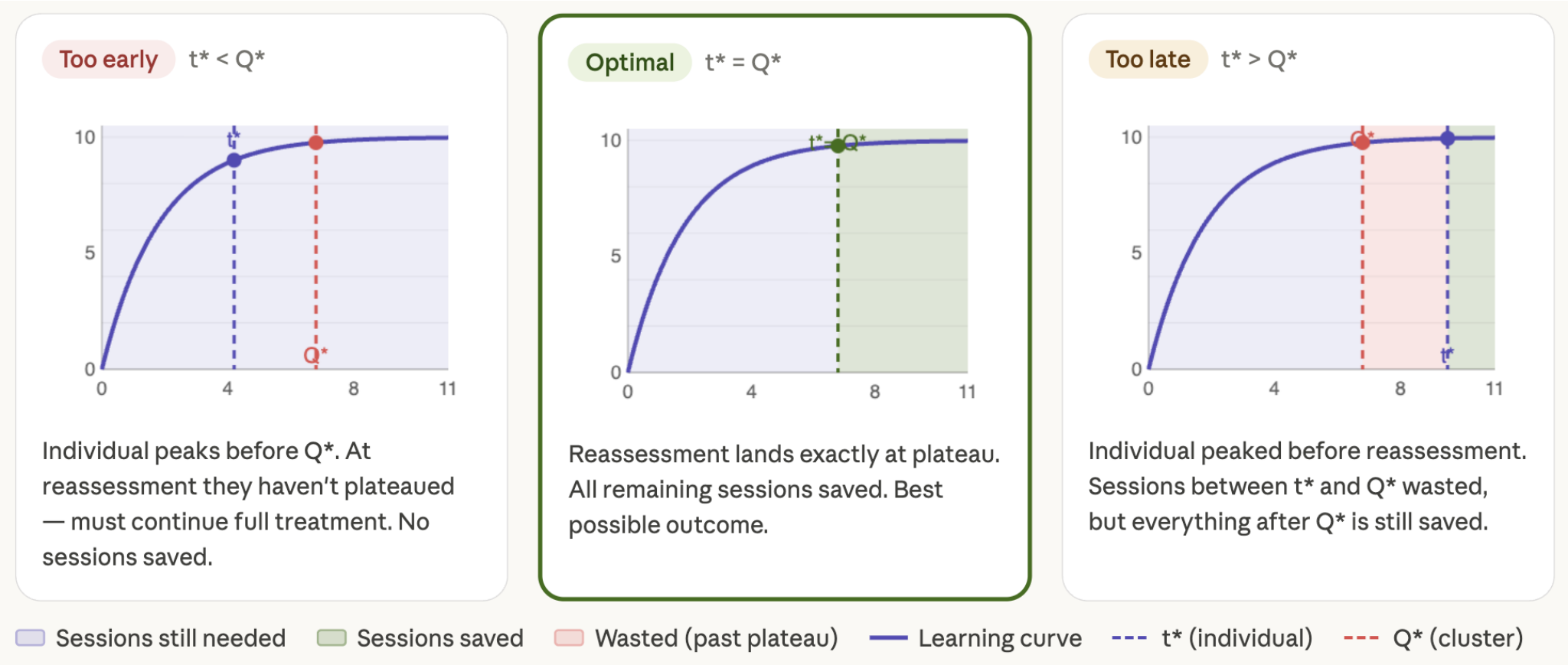
- In a classic M/M/1 queue, a “customer” arrives, receives one service interaction, and departs (immediately freeing up capacity)
- Rehabilitation services is multi-appointment: a client receives a **sequence of sessions** (referred to as a treatment block)
- The **Break for Practice** gives families time to practice and generalize skills in daily life so the clinician can reassess later and decide whether to continue, adjust, or discharge



UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

The Reassessment Trade-off

- If we assume there's only one opportunity to stop and reassess, then we have three scenarios based on where Q* falls relative to each client's t* within that cluster



UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Mathematical Formulation

1. One-shot assessment setup

For each cluster c, each child i has a true stopping point t_i^*. We choose a single policy Q_c : one reassessment point for all children in cluster c.

$\mathbf{t_i^*}$ true plateau for child i in cluster c

$\mathbf{Q_c}$ policy: one reassessment point for cluster c

2. Sessions delivered (cost)

$$Delivered(Q, t_i^*) = \begin{cases} T_{max} & \text{when } Q < t_i^* \text{ (assessed too early, no savings)} \\ Q & \text{when } Q >= t_i^* \text{ (assessed late enough, saves } T_{max} - Q) \end{cases}$$

Goal: choose Q_c to minimize expected sessions delivered

3. Capacity planning objective

$$Max_Q \;\; E[\, savings(Q, t^*) \,]$$

Equivalently: min_Q E[delivered(Q,t*)]. Choose Q to maximise expected sessions saved across the cluster.

4. Expected savings

$$savings(Q, t_i^*) = \begin{cases} 0 & \text{when } Q < t_i^* \\ T_{max} - Q & \text{when } Q >= t_i^* \end{cases}$$

$$E[savings](Q) = P(t^* \leq Q) \cdot (T_{max} - Q)$$

Since savings are non-zero only when Q >= t*, the expectation reduces to CDF × sessions saved.

5. Optimal policy Q*

$$Q^*_c = arg\ max_Q \;\; [\, F_c(Q) \cdot (T_{max} - Q) \,]$$

$\mathbf{F_c(Q)}$ = probability that a randomly drawn child from cluster c has already plateaued by Q

UNIVERSITY OF WATERLOO | FACULTY OF ENGINEERING

Policy Implications

- **Use Q* as the default reassessment point**
  - For each cluster, Q* provides an evidence-based session at which to schedule reassessment. It minimizes expected wasted sessions under uncertainty
- **Cluster assignment drives the policy**
  - When a new client is referred, assign them to the most likely cluster based on early sessions. Their cluster's Q* becomes the planned reassessment date.
- **Within-cluster variance sets uncertainty bounds**
  - The spread of individual t* values around Q* defines a confidence window. Wider spreads may warrant earlier reassessment with a follow-up protocol.

Takeaways: Capacity/Scheduling Optimization

1. Clustering by trajectory reveals systematic patterns that cannot be seen at the individual level
2. Q* is the cluster-level **α %** plateau
   - This serves as a cluster's reassessment point
3. Alpha is a tunable threshold that defines t* and Q*
4. Individual t* values vary around Q* within each cluster
   - This spread determines policy uncertainty
5. Reassessing too early is worse than too late: early reassessments save no sessions, late reassessments save sessions beyond Q*
6. Cluster assignment at referral allows proactive scheduling of the reassessment session